sas innovate 2026

Hands-on Workshop:

Generating Synthetic Data Using SAS Data Maker

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SAS

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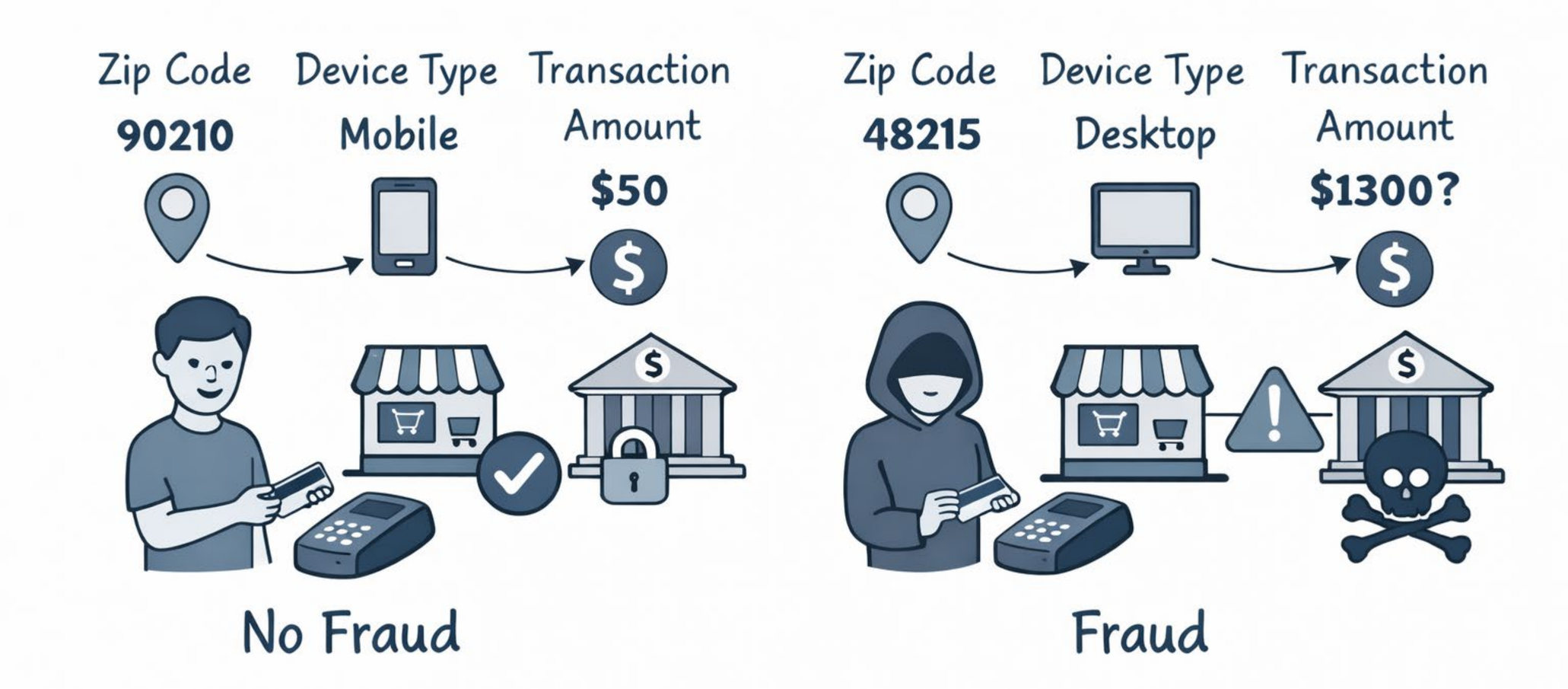
Fraud Data Example



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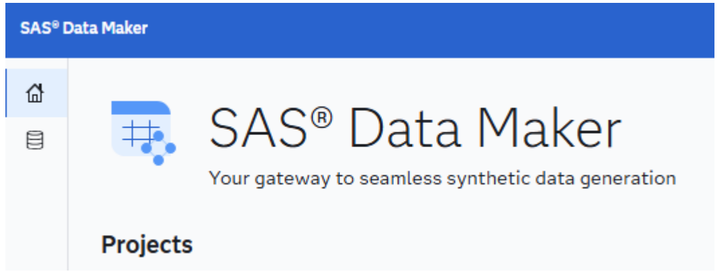
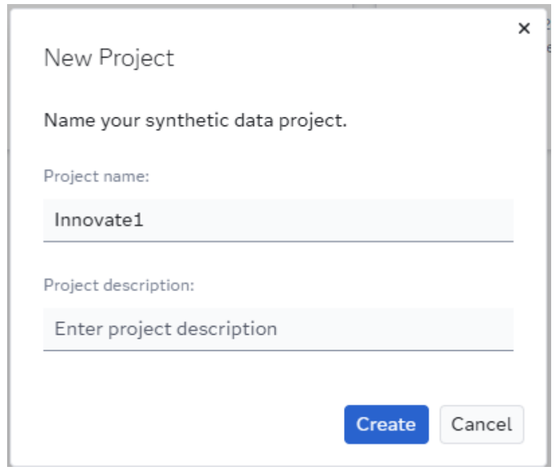
Create a Project in SAS Data Maker





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Why Synthetic Data?

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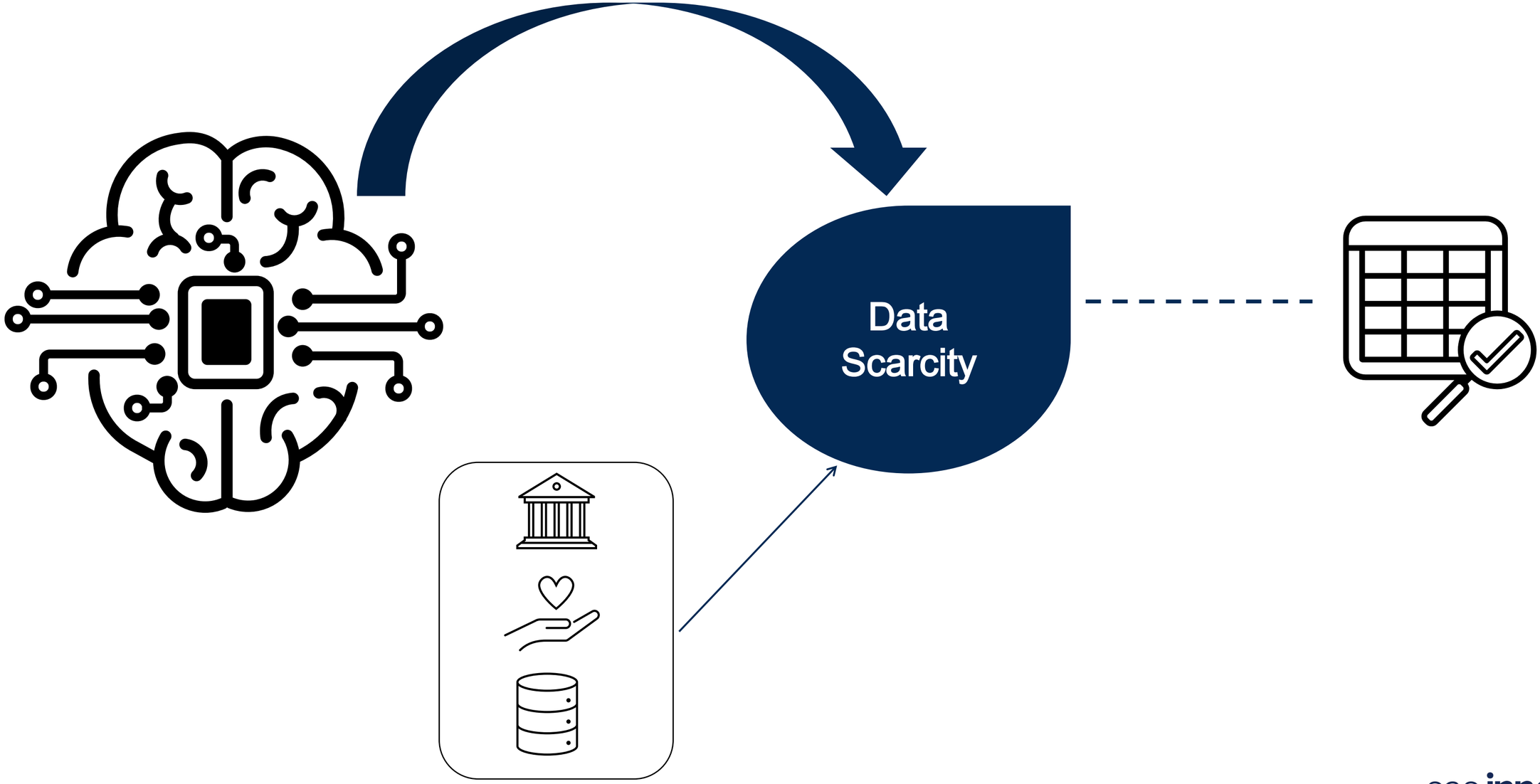
The Data Problem



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Challenges in Gathering Suitable Data

Sensitive or private information	Internal or external privacy controls can restrict the access, usage and sharing of real data
Prohibitive cost	The cost to purchase or collect real data can be prohibitive.
Bias or imbalances	Sufficient representation among groups or segments, especially sensitive variables, can be lacking.
Manual labeling inefficiency	Labeling real data manually is time-intensive and error-prone

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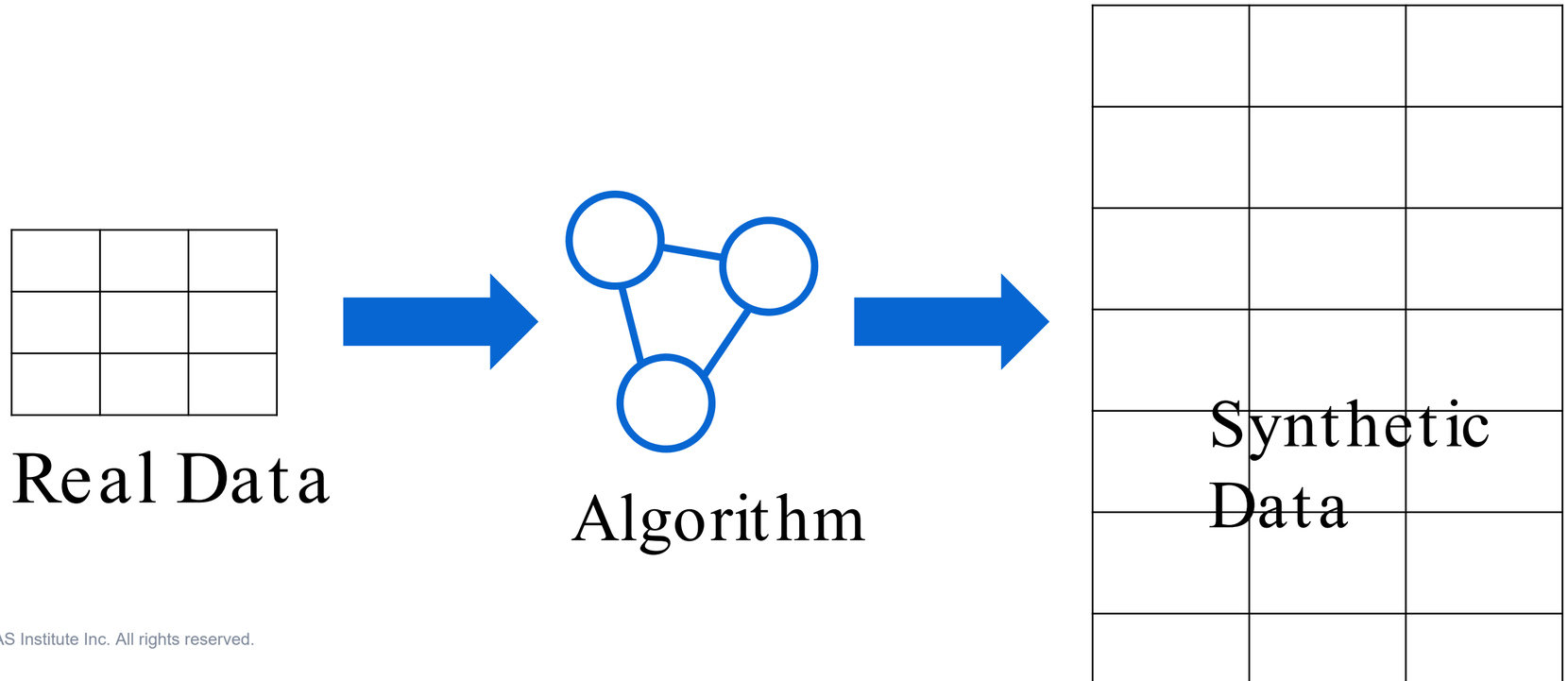
The Solution

Synthetic Data

Synthetic data is algorithmically generated data that mimics real-world data



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How Synthetic data can help?

Privacy Preserving	Does not expose real, identifiable information.
Cost Effective	Cost-effective alternative to manual processes for collecting real-world data
Balanced	Addresses data bias challenges by utilizing diverse data augmentation and generation methods
Scalable	Easier to create and offers flexibility, scalability and addresses data scarcity.

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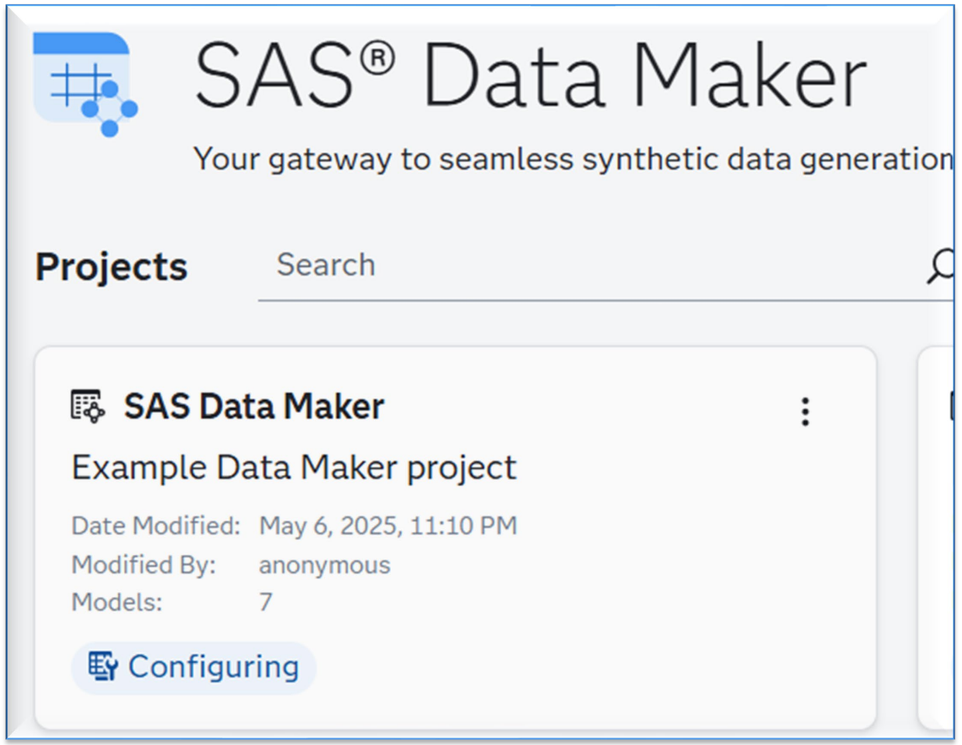
The SAS Data Maker

SAS Data Maker

Generate Synthetic Data

Mitigate Data Quality and Scarcity



Available Methods

- ✓ Bayesian Network
- ✓ SMOTE

A low-code/no-code interface to augment or generate data quickly

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SAS Data Maker Capabilities

Address gaps and limitations in real-world data

1

Generate data on demand

2

Replicate data characteristics

3

Evaluate synthetic data quality

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SAS Data Maker Process

Synthetic data generation follows these common steps

Plan	Prepare	Produce
01 Onboard data	04 Configure generation parameters	06 Assess generated data
02 Select variables	05 Train generation model	07 Generate synthetic data
03 Pre-process original data		08 Curate and release data

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Methods for Generating Data

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Methods for Generating Synthetic Data

Rule-Based

Use predefined rules to create data with specific characteristics

Statistical

Addressing imbalance in rare categories by creating artificial samples

AI-Driven

Creating artificial data that resemble real data for tabular, text, image and video data generation

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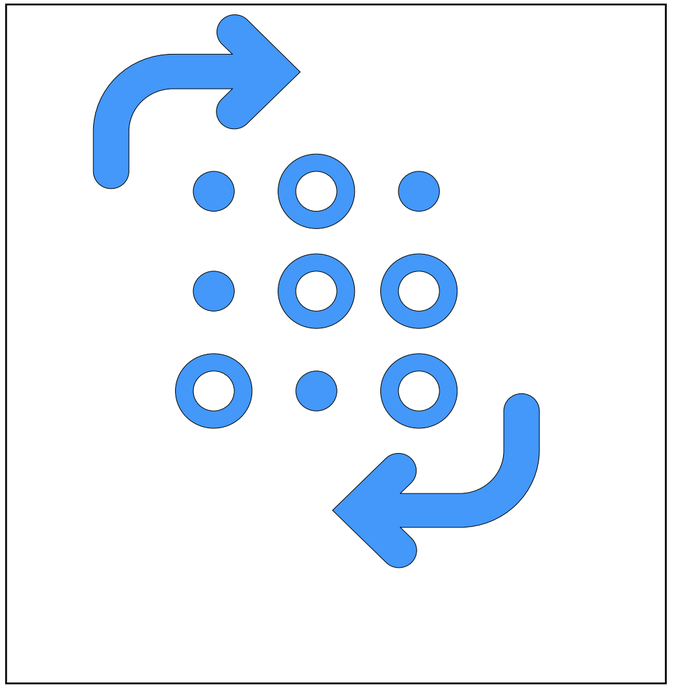
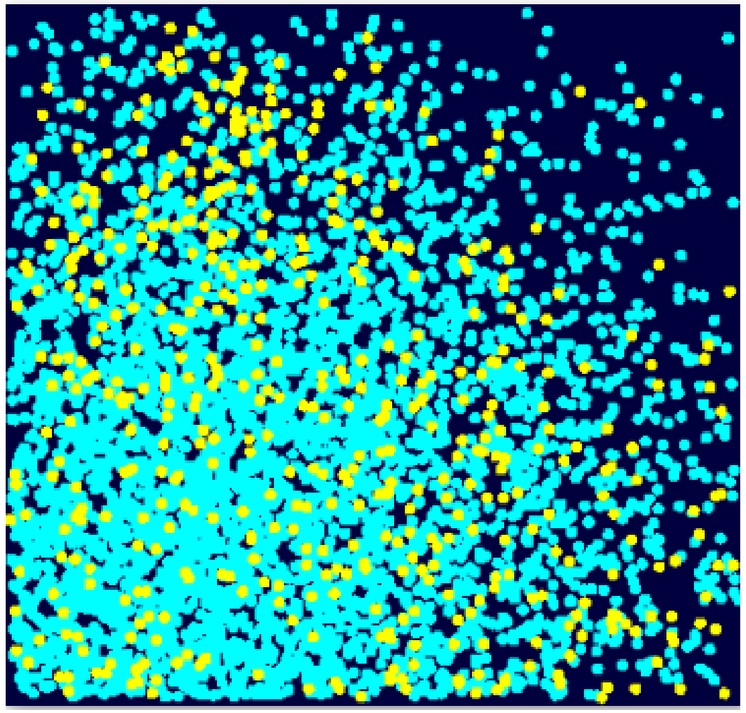
Common Techniques for Generating Synthetic Data

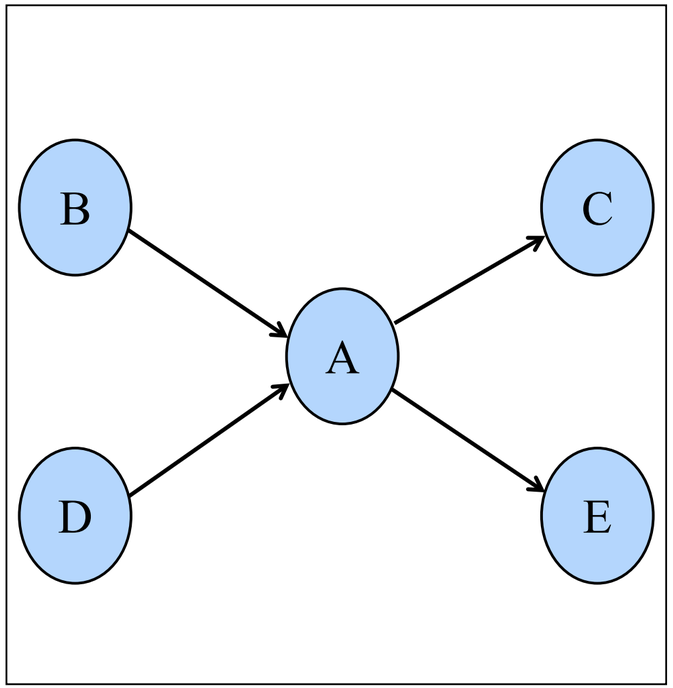
Generative Adversarial Networks (GANs)



Synthetic Minority Oversampling Technique (SMOTE)



Bayesian Network



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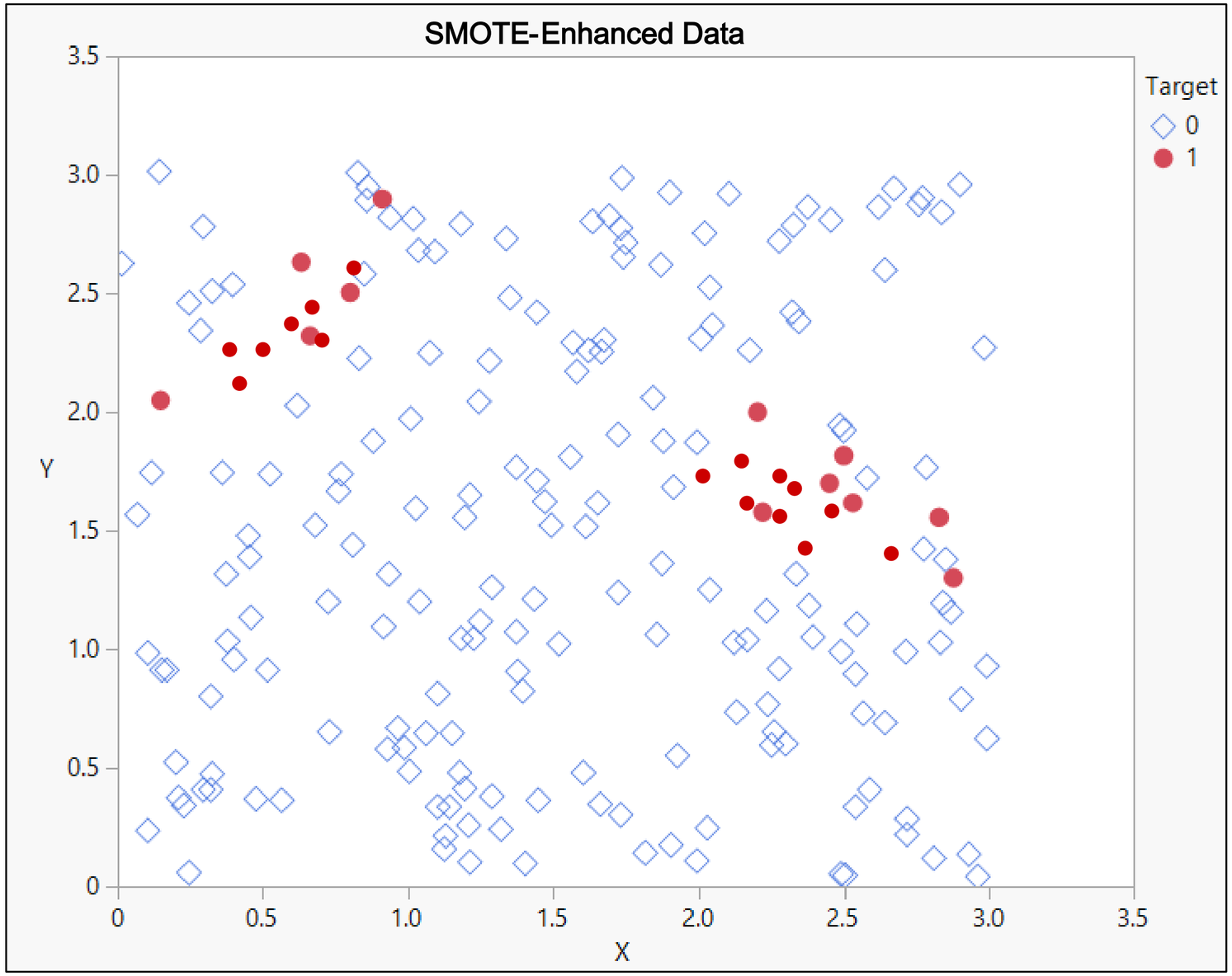
What is SMOTE



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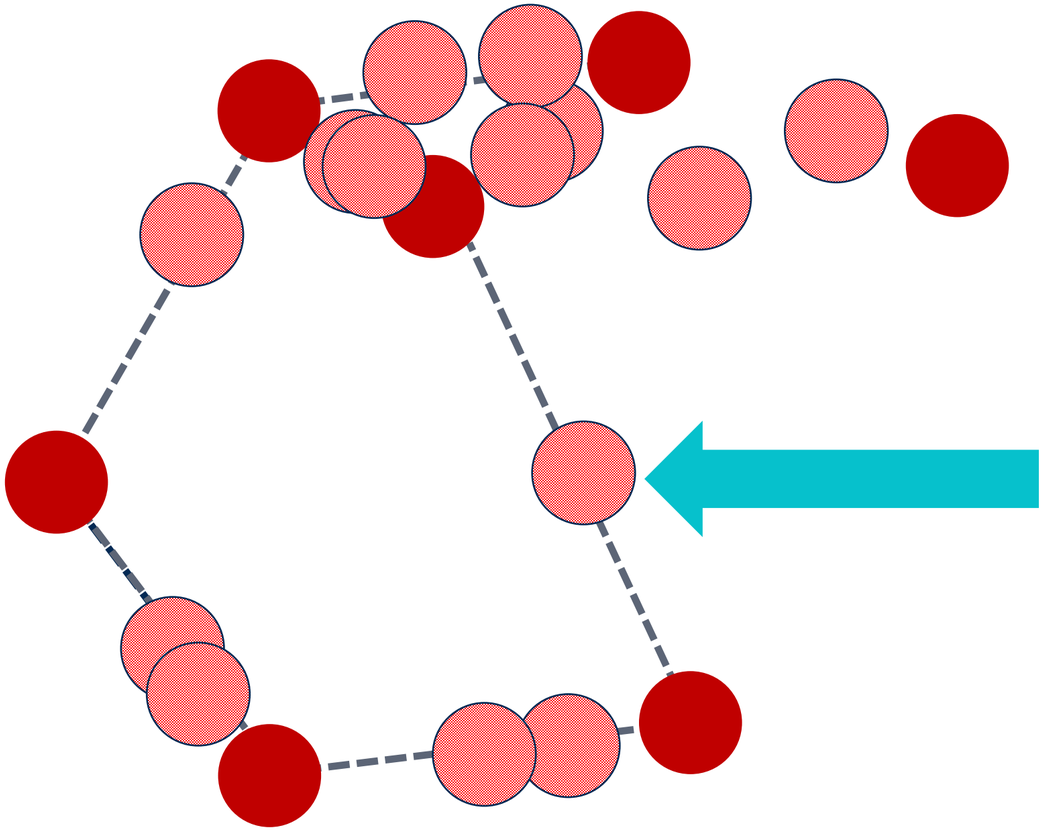
How SMOTE Works (only rare -event cases)

example: k=2

threshold



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Set Up the Columns

Structure | Columns

Table	Column	Semantic type	Entity	Valid	Included
credit_fraud_events	AccountTenureMonths	Integer		✓	
credit_fraud_events	AvgTransactionAmountLast3...	Float		✓	
credit_fraud_events	CardType	Category		✓	
credit_fraud_events	ChargebackWithin30Days	Category		✓	
credit_fraud_events	CustomerAge	Integer		✓	
credit_fraud_events	CustomerRegion	Category		✓	
credit_fraud_events	DeviceType	Category		✓	
credit_fraud_events	Flag	Constant		✓	
credit_fraud_events	HasPriorFraud	Category		✓	
credit_fraud_events	HashedCustomerID	(Not set)		✓	⊘
credit_fraud_events	IsHighRiskCountry	Category		✓	
credit_fraud_events	IsInternational	Category		✓	
credit_fraud_events	IsOnline	Category		✓	
credit_fraud_events	MerchantCategory	Category		✓	
credit_fraud_events	MerchantCountry	Country	location 49	✓	
credit_fraud_events	MerchantName	Category		✓	

New Project

Name your synthetic data project.

Project name:

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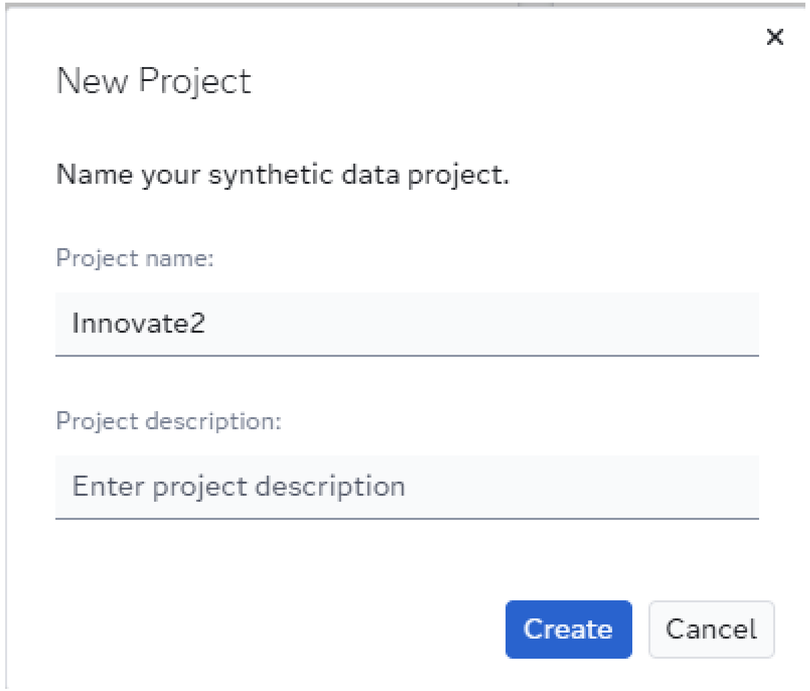
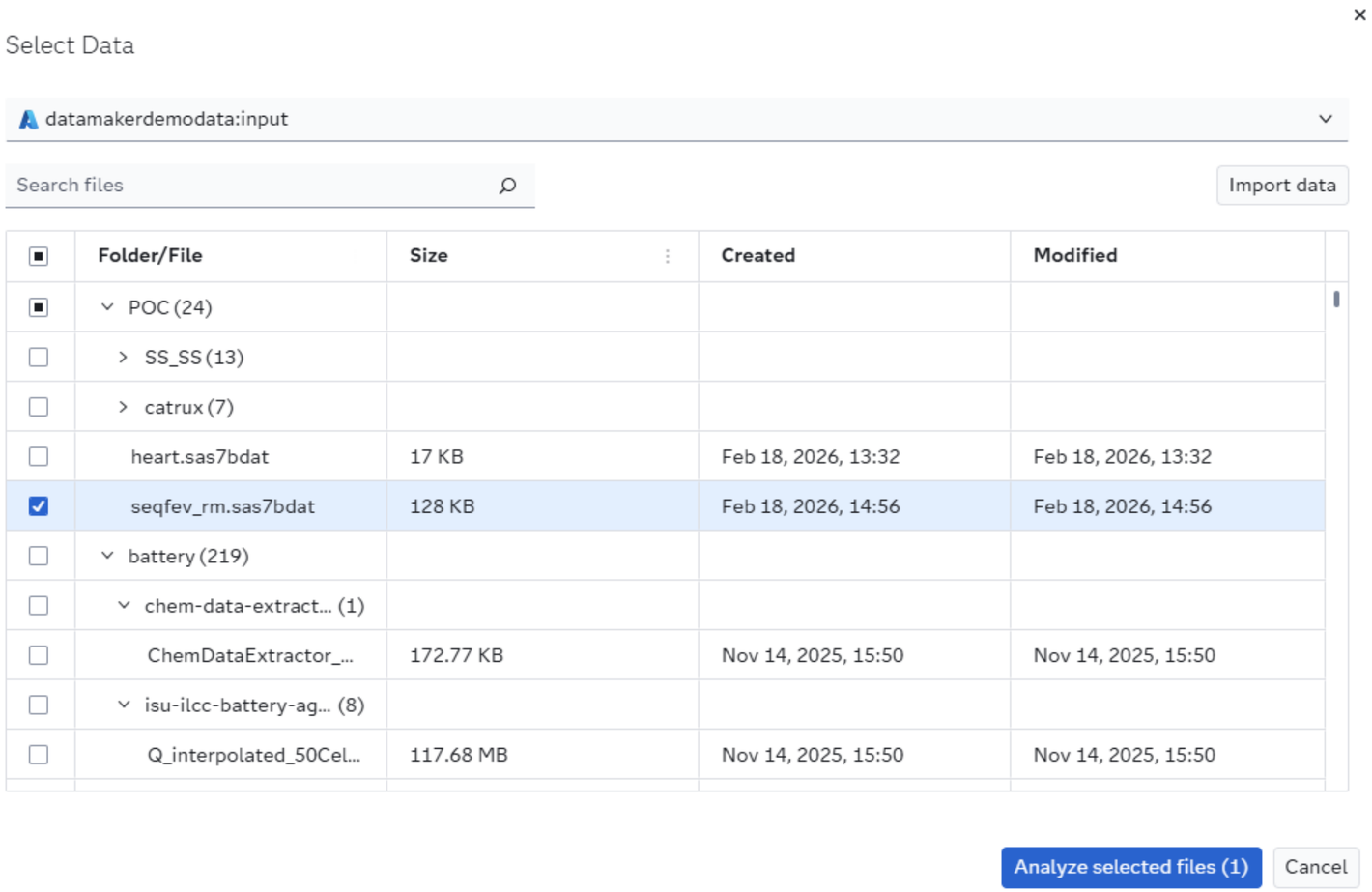
Create a New Project





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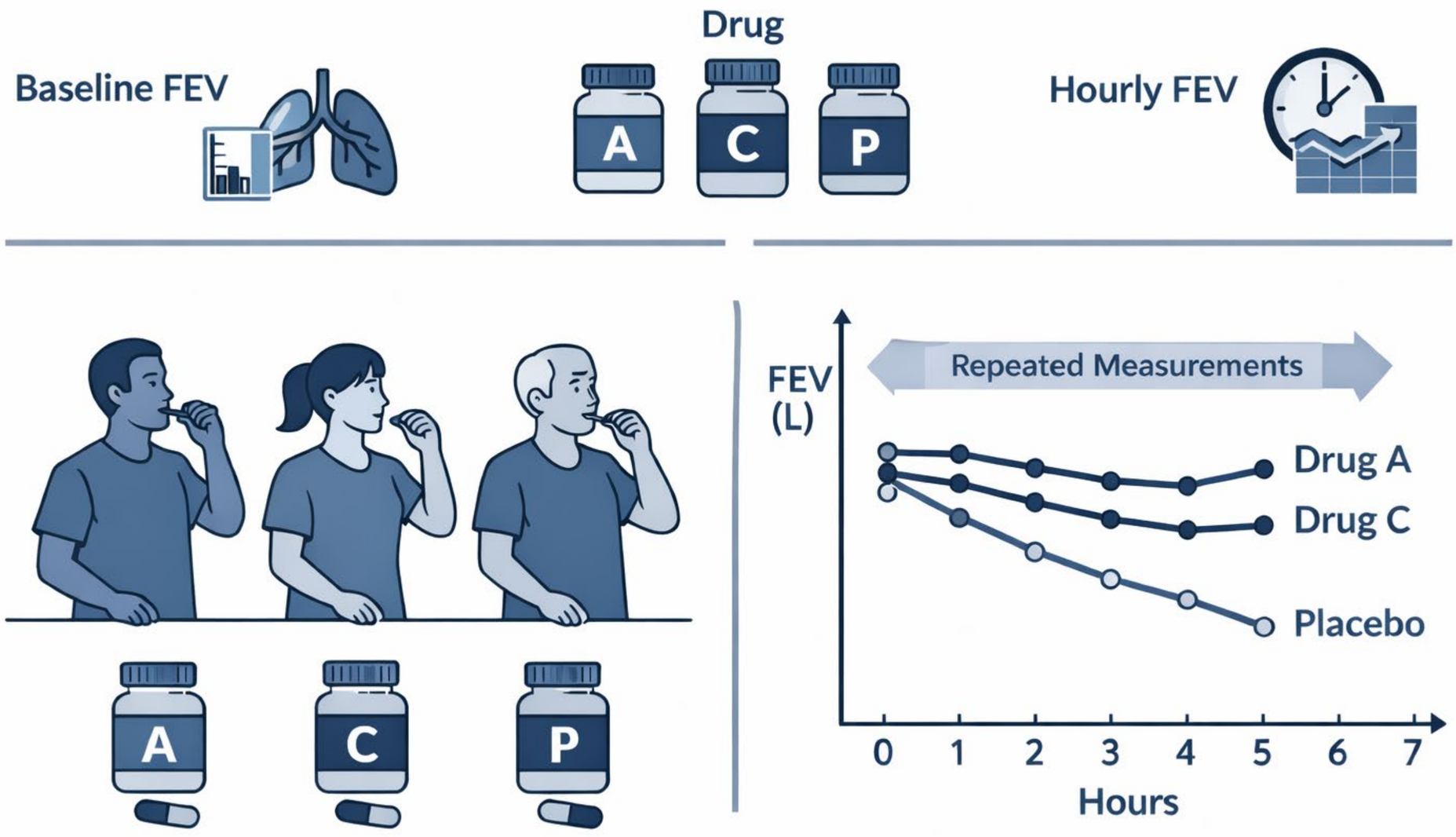
Asthma Repeated Measures Example



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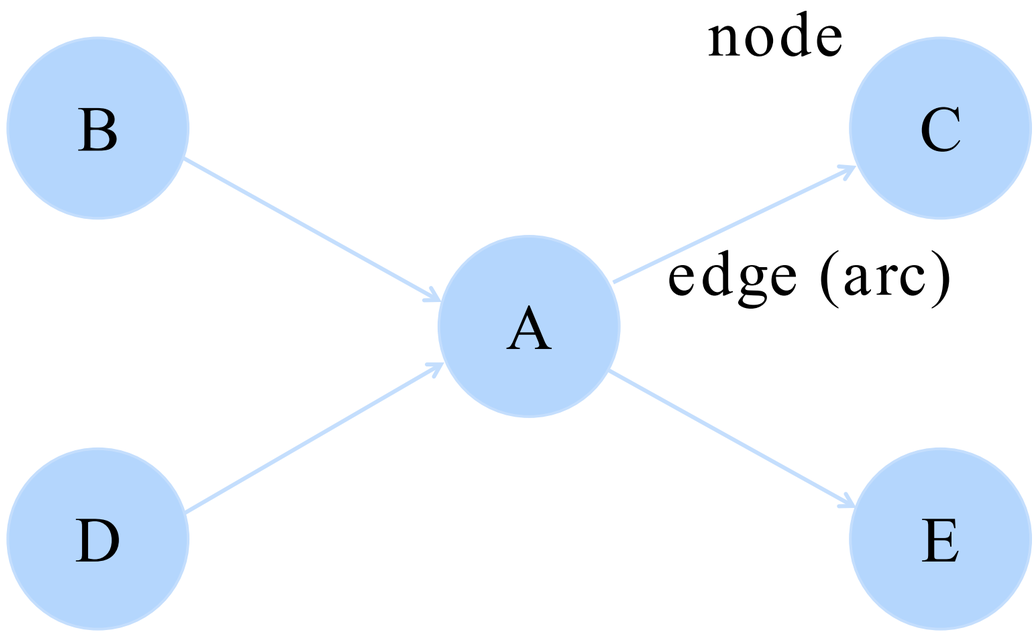
Bayesian Network

A Bayesian network is a graphical model that consists of two parts:

- a directed acyclic graph (DAG) in which nodes represent random variables and the links between the nodes represent conditional dependency between random variables
- a set of conditional probability distributions, one for each node conditional on its parents



directed acyclic graph (DAG)

- node = random variables
- edge = conditional dependency between random variables

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Reference:
How Data Maker Works

Phase 1: Plan & Prepare

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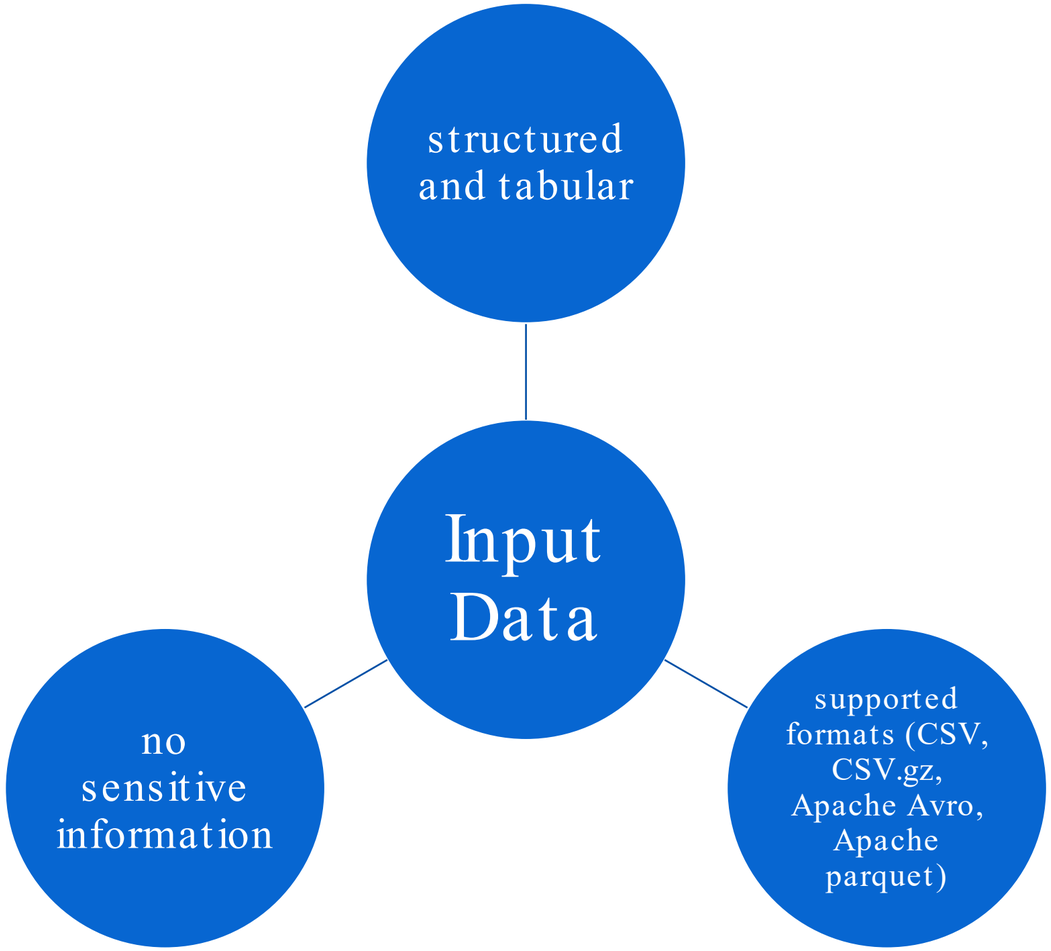
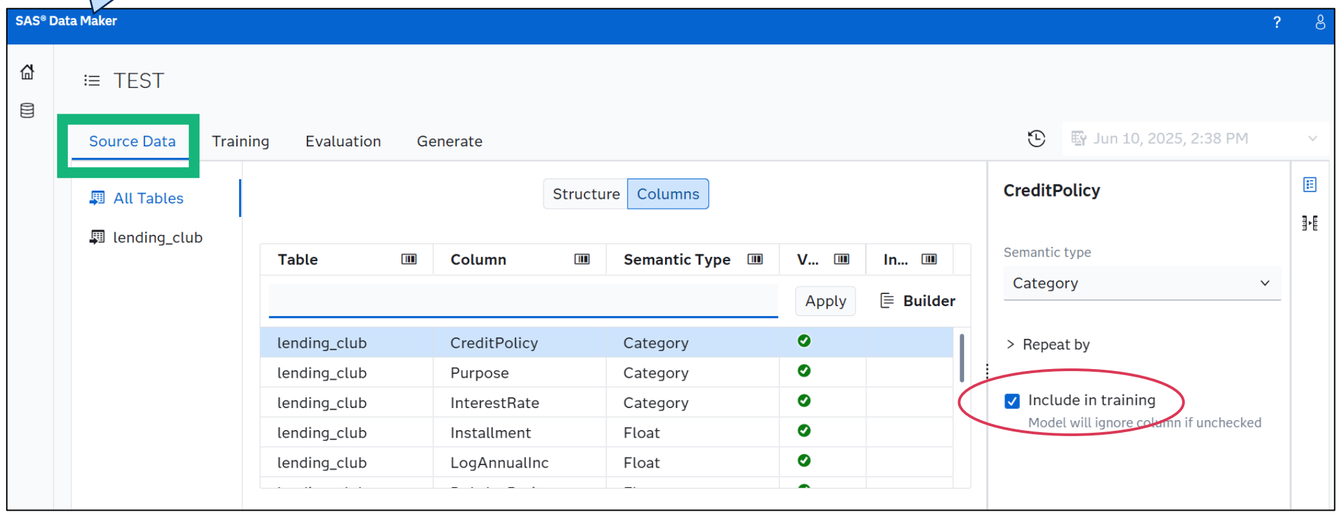
SAS Data Maker Process



Allows you to select multiple data sources

Plan



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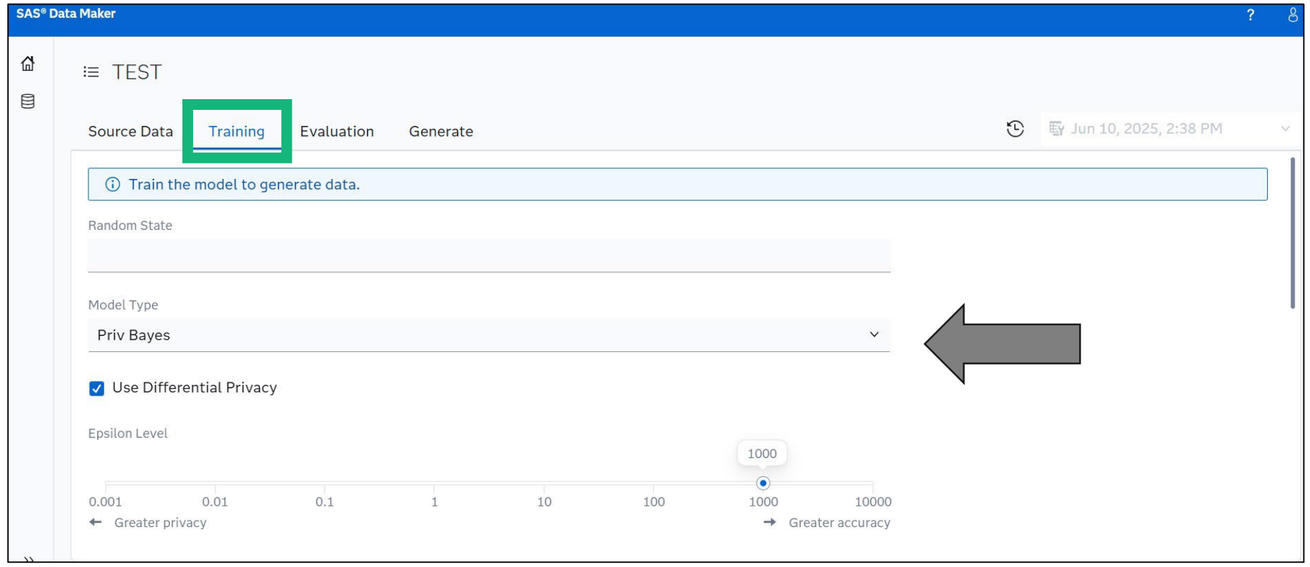
SAS Data Maker Process

Prepare

Bayesian Network Method	SMOTE Method
• Structured • Tabular	• Structured • Tabular • Non-temporal



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Reference:
How Data Maker Works

Phase 2: Evaluate

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SAS Data Maker Process

Produce

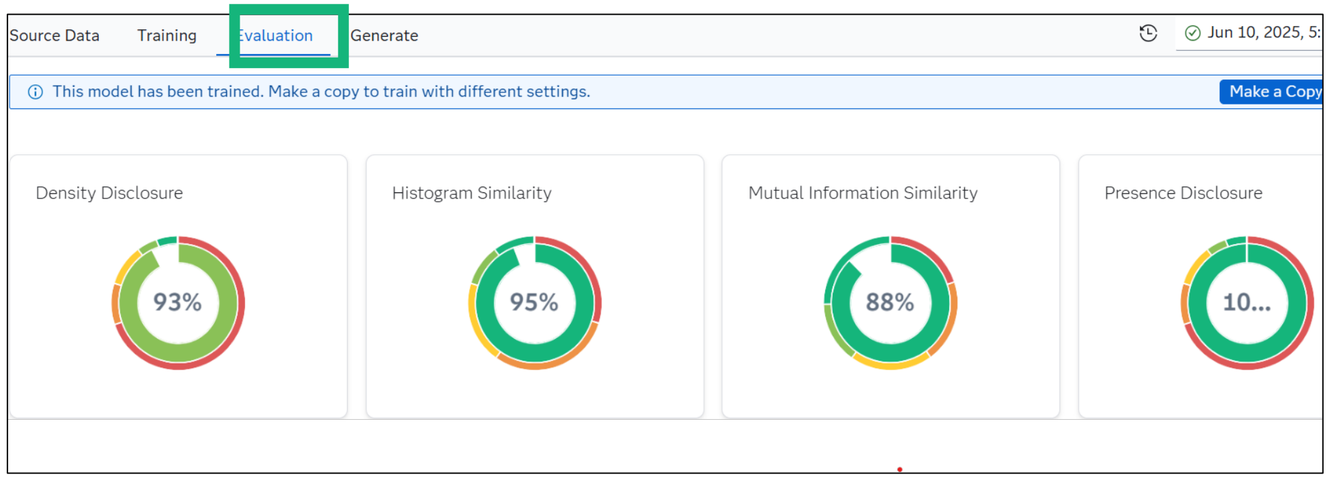
Assessment Metrics

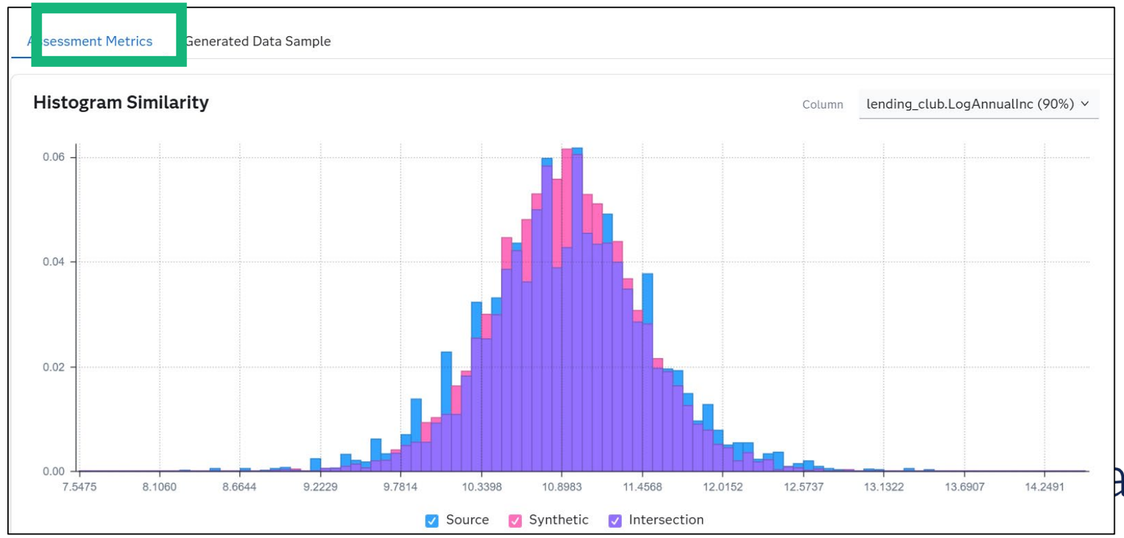
Similarity Metrics

- Histogram Similarity
- Mutual Information Similarity
- Cross-Table Mutual Information Similarity
- Degree Distribution Similarity

Privacy Metrics

- Density Disclosure
- Presence Disclosure





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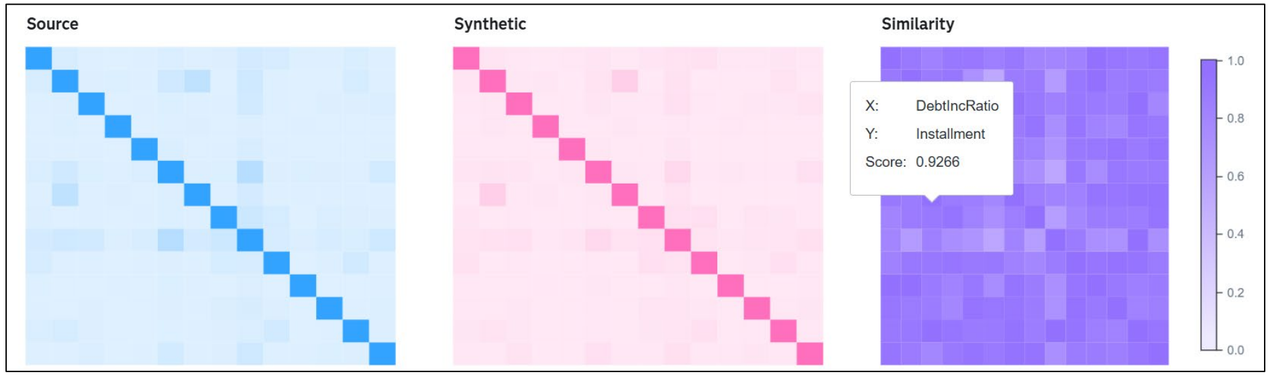
SAS Data Maker Process

Similarity Metrics

Measures how closely synthetic data replicates the patterns, distributions, and relationships present in the source data

- Histogram Similarity
- Mutual Information Similarity
- Cross-Table Mutual Information Similarity
- Degree Distribution Similarity



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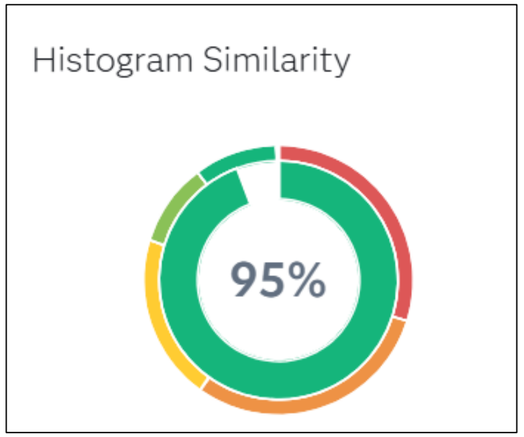
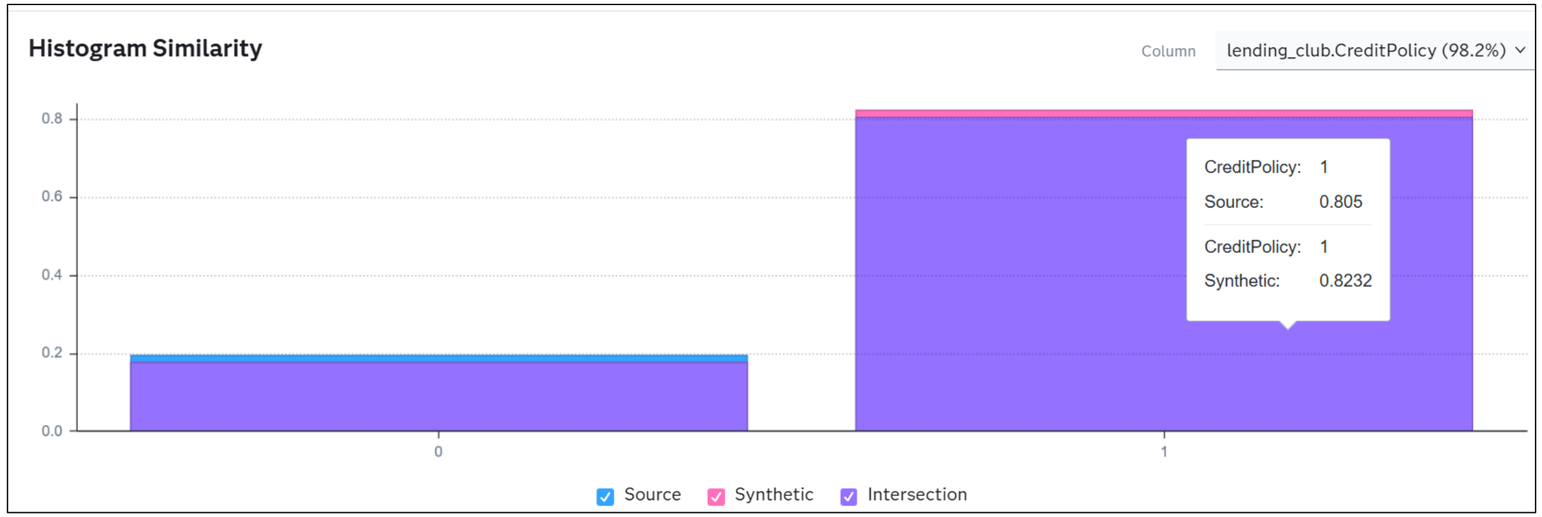
SAS Data Maker Process

Similarity Metrics

Histogram Similarity

- evaluates how well the marginal distribution of each column is preserved in the synthetic data
- outer ring in histogram similarity gauge represent different grade ranges for a score that ranges between 0 and 1
- histogram measures the percent of observations that occur in a bin range in both the original source input data and the generated synthetic data





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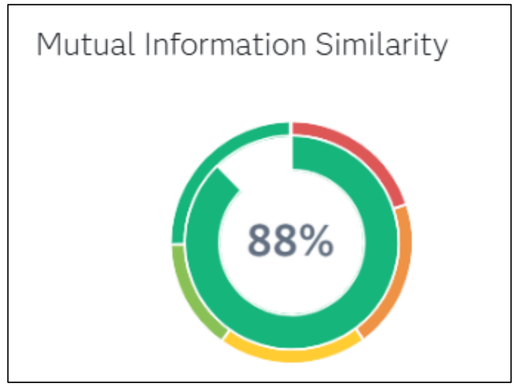
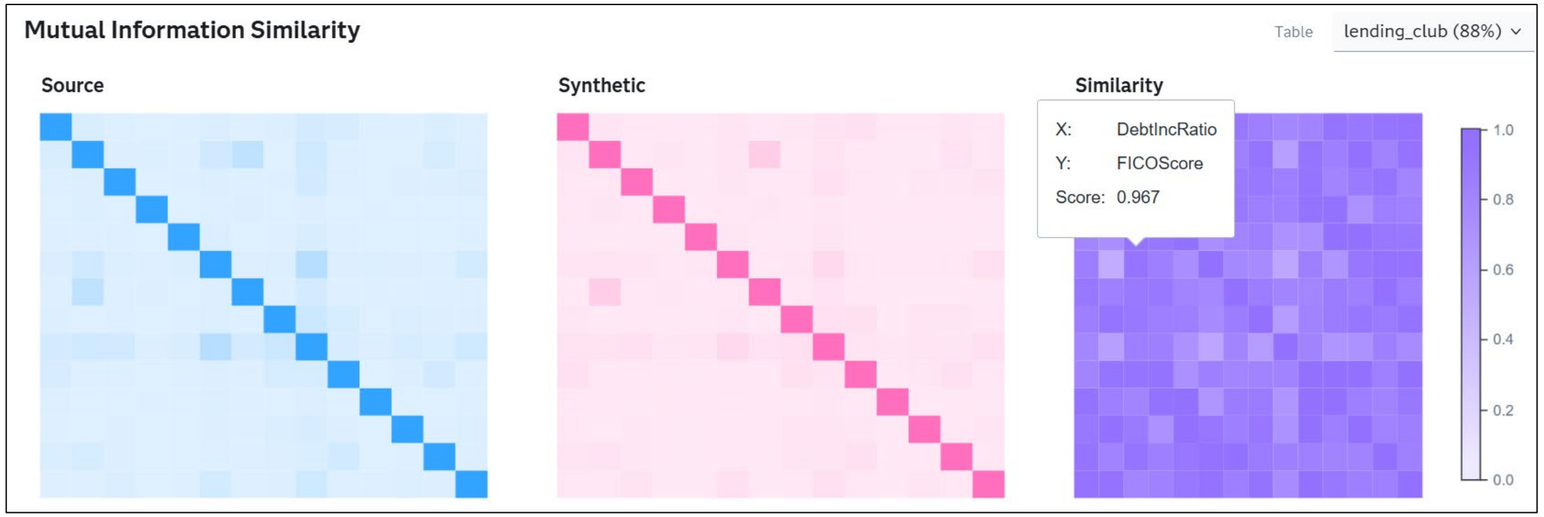
SAS Data Maker Process

Similarity Metrics

Mutual Information Similarity

- measures the mutual dependence between the two variables
- recommended when you need to verify that the relationship between specific columns (such as “height” and “weight”) is preserved
- outer ring in mutual information similarity gauge represent different grade ranges for a score that ranges between 0 and 1





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SAS Data Maker Process

Similarity Metrics

Cross-Table Mutual Information Similarity

- used in a multi-table context
- evaluates how effectively models preserve relationships between tables
- similar to the standard mutual information similarity metric with the additional constraint the pair of columns must be in different tables.

Degree Distribution Similarity

- metric is meaningful only in a multi-table setting
- evaluates how well the connectivity between related tables (one-to-many and many-to-many) is preserved between the source data and the synthetic data

☞ Summary percentage, histograms or bar charts and score for all columns are also displayed as discussed earlier

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SAS Data Maker Process

Privacy Metrics

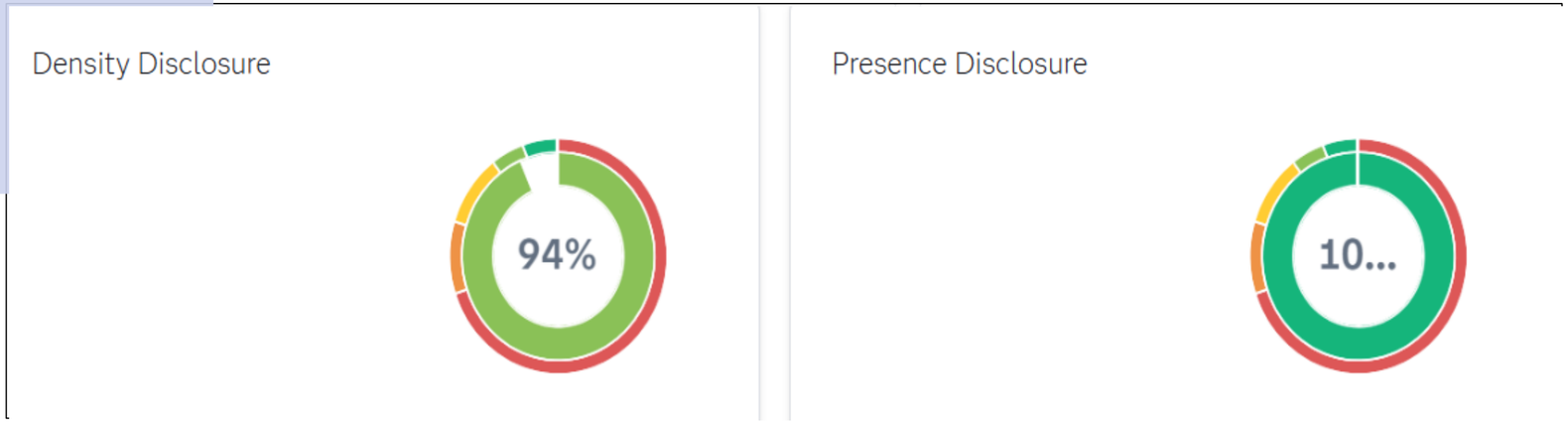
Measures the risk of identifying a user in the database.

Recommended whenever the data is to be shared with external third-parties or any untrusted group or individuals

- Density Disclosure
- Presence Disclosure

Anonymization

Re-identification



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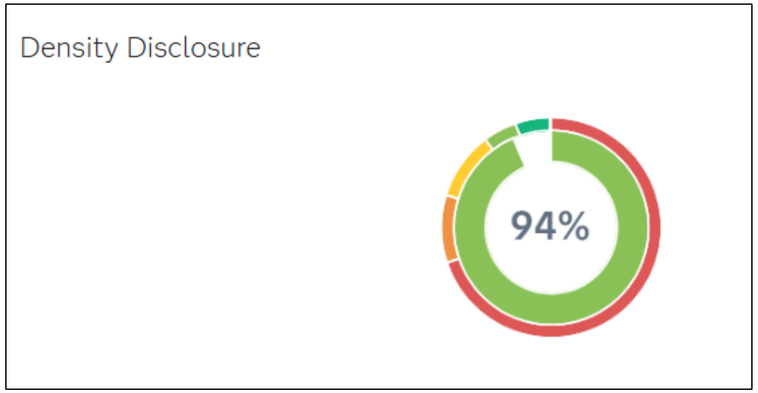
SAS Data Maker Process

Privacy Metrics

Density Disclosure

- estimates the risk that an adversary could potentially construct a mapping between some synthetic data points and real data points.
- outer ring in density disclosure gauge represent different grade ranges for a score that ranges between 0 and 1.



Values closer to 1

- Indicates low disclosure risk
- Synthetic data is very similar to the original in a certain neighborhood

Values closer to 0

- Indicates high disclosure risk
- Synthetic data is sufficiently different from the original in a certain neighborhood

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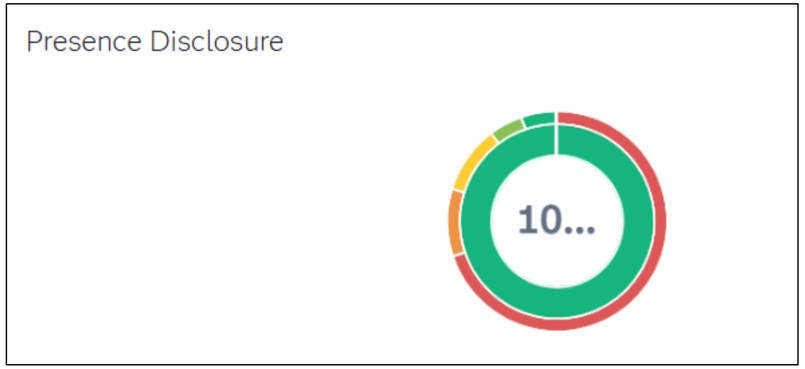
SAS Data Maker Process

Privacy Metrics

Presence Disclosure

- estimates the risk that an adversary could infer whether an arbitrary data point was part of the training set of a synthetic data generator.
- outer ring in density disclosure gauge represent different grade ranges for a score that ranges between 0 and 1.



Values closer to 1

- Indicates low presence disclosure risk
- Unlikely to infer whether a given data point was in the training set

Values closer to 0

- Indicates high presence disclosure risk
- Features in the synthetic data can determine which records were in the training set

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Reference:
How Data Maker Works

Phase 3: Generate

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SAS Data Maker Process

Generate Synthetic Data

Produce

Source Data | Training | Evaluation | Generate

This model has been trained. Make a copy to train with different settings.

No generated data to show

SOURCE DATA

Tables	Columns	Rows	Size
1	14	9.6K	1.02 MB

GENERATED DATA

Tables	Columns	Rows
--	--	--

Location: N/A

Create your synthetic data set.

Generate Synthetic Data

Generate Synthetic Data

Output destination

Output path

Output format: csv

Random state

Output rows: 9578

Output table: lending_club

Or

Magnitude: 1

Generate | Cancel

SOURCE DATA

Tables	Columns	Rows	Size
1	14	9.6K	1.02 MB

GENERATED DATA

Tables	Columns	Rows
1	14	2K

Location: N/A

Column Types

Table Types

Schema

Schema				Synthetic samples	
Table	Column	Type	Distribution	Sample #1	Sample #2
lending_club	CreditLineAge	Float		2605.17	6169.59
lending_club	CreditPolicy	Category		1	1
lending_club	DebtIncRatio	Float		14.43	6.01
lending_club	Default	Category		0	0
lending_club	Delinquencies2...	Category		0	0
lending_club	FICOScore	Category		727	787
lending_club	Inquiries6Mnths	Category		0	0
lending_club	Installment	Float		326.67	143.06
lending_club	InterestRate	Category		0.1114	0.074

Profile: Generated Data

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Thank you!

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